Green infrastructure for stormwater management: A geospatial framework separating feasibility screening from multicriteria prioritization

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Abstract

Nature-based solutions, including green infrastructure, can support stormwater management by reducing runoff, increasing storage, and enhancing infiltration. However, the benefits of individual best management practices (BMPs) depend on where they can be sited relative to runoff sources and hydrological conditions. This study develops Prioritization of Locations After Constraint Evaluation (PLACE), a geospatial framework that explicitly separates BMP-specific feasibility screening from prioritization within the resulting feasible domain. This distinction preserves the difference between locations that fail required siting conditions and locations that remain feasible but differ in relative priority. PLACE proceeds through three sequential steps of existing-condition characterization, BMP-specific binary feasibility screening, and multicriteria prioritization within each feasible domain using Analytic Hierarchy Process-derived weights. Event-based runoff and storage estimation and sensitivity analysis are used to estimate screening-level retention potential and evaluate priority stability. The framework is demonstrated in Bonita Bay, Florida, for 11 BMP types across three runoff sources of roofs, parking/roads, and pervious areas. Results show that at-source practices have broader feasible domains, whereas BMPs requiring adjacent or residual pervious land are restricted by fragmented space and hydrological constraints. Priority varies within feasible domains, indicating that feasibility alone cannot identify where subsequent evaluation should be concentrated. Results also show that screening-level retention potential is highest where runoff-generating surfaces coincide with extensive feasible BMP domains, as demonstrated by pervious pavement in parking/road areas. Priority patterns remain stable under ±10% perturbations of criterion weights. PLACE provides an adaptable screening framework for narrowing and prioritizing locations for subsequent hydrological, hydraulic, and site-specific evaluation.

Keywords: Nature-based solutions; green infrastructure; stormwater best management practices; geospatial multicriteria analysis; hydrological prioritization; stormwater management

1. Introduction

The hydrological responses of urban watersheds are altered by increasing impervious surface coverage, which reduces infiltration and increases stormwater runoff and peak discharge (Iqbal et al., 2025; Li et al., 2019; Ongaga et al., 2024; Öztürk et al., 2024; Samborska-Goik et al., 2025). These changes can increase flood risk and associated economic losses (Agonafir et al., 2023; Xing et al., 2021), while disruption of natural infiltration pathways can reduce groundwater recharge (Pasquier et al., 2022; Šiljeg et al., 2023). Runoff from impervious surfaces can also transport contaminants to surface-water and groundwater systems, affecting water quality and ecosystem health (Motlagh, 2025). Conventional gray stormwater infrastructure, including engineered conveyance and storage systems, has finite design capacity and may become increasingly stressed as urbanization and hydrological extremes intensify (Lu & Wang, 2021). Expanding conveyance capacity alone may also transfer stormwater and its associated impacts downstream rather than managing runoff near its source (Zellner & Massey, 2024). These limitations motivate stormwater-management strategies that complement conventional infrastructure by managing runoff closer to where it is generated. Nature-based solutions including green infrastructure (GI) provides one such strategy by using vegetation, soils, storage, and infiltration processes to manage stormwater runoff (Adhikari et al., 2024; Ahiablame et al., 2012; Baldwin et al., 2022; Faruk et al., 2025; Wang et al., 2025; Wang et al., 2024). In this study, GI refers to the broader stormwater-management strategy, whereas a best management practice (BMP) refers to an individual stormwater intervention implemented within that strategy. Depending on the type of practice, BMPs can store runoff, promote infiltration, and reduce peak discharge. However, the potential benefits of BMPs depend on their placement relative to runoff sources and local site conditions (Martin-Mikle et al., 2015; Tsegaye et al., 2019).

Recent studies have increasingly emphasized the spatial placement of BMPs and their relationship to hydrological outcomes (Adhikari et al., 2024; Ambily et al., 2025; Arteaga-Zambrano et al., 2025; Caroppi et al., 2024; Roozbahani et al., 2025; Wang et al., 2023). Process-based hydrological and hydraulic models can provide detailed assessments of BMP performance, but their data and computational requirements can limit their practicality for the initial screening of numerous BMP alternatives and candidate locations (Liu et al., 2023; Sehrawat & Shekhar, 2025; Zellner & Massey, 2024). Accordingly, geospatial analysis approaches have been widely used to support BMP screening by integrating environmental, hydrological, and land-surface information into spatial site selection analyses (Aghaloo & Sharifi, 2025; Allafta & Opp, 2021; Goyal & Biswas, 2025; Liu et al., 2023).

Geospatial analysis approaches commonly combine factors such as slope, soil, land use, drainage characteristics, and proximity or buffer relationships to identify potentially suitable locations for stormwater BMP interventions (Harlis & Seo, 2024). However, when prerequisite feasibility constraints and indicators of potential hydrological benefit are incorporated into a single suitability score, the distinction between feasibility and priority can become unclear. A location that satisfies the minimum requirements for a BMP does not necessarily have high priority, while a high score on one criterion should not compensate for failure to satisfy a prerequisite constraint. Accordingly, geospatial-based approaches can provide an initial screening and ranking of candidate stormwater-management locations before detailed site investigation or hydrological simulation (Inamdar et al., 2013; Kaykhosravi et al., 2022). BMP planning can therefore be framed as two sequential spatial questions of where a particular BMP can be implemented and which of those feasible locations should receive higher priority.

This study addresses these two questions through the Prioritization of Locations After Constraint Evaluation (PLACE) framework, a geospatial framework that explicitly separates feasibility from priority (Figure 1). In the PLACE framework, feasibility indicates whether a location satisfies all applicable BMP-specific feasibility constraints, such as location requirements, proximity to a runoff source, slope, groundwater separation, and required buffers. These constraints are treated as binary requirements, such that a location that fails any applicable constraint is excluded from the feasible domain for that BMP. Priority is evaluated only after feasibility has been established and represents the relative ranking of locations within the feasible domain according to priority criteria. In this study, these criteria represent runoff generation, hydraulic conductivity, storage adequacy, distance to natural drainage, and terrain suitability. Thus, feasibility constraints determine where a BMP can be implemented, whereas priority criteria determine which feasible locations should be considered first. In other words, a location must satisfy the BMP-specific feasibility requirements before multicriteria prioritization is performed. This sequential structure prevents high scores on priority criteria from compensating for failure to satisfy a prerequisite feasibility constraint.

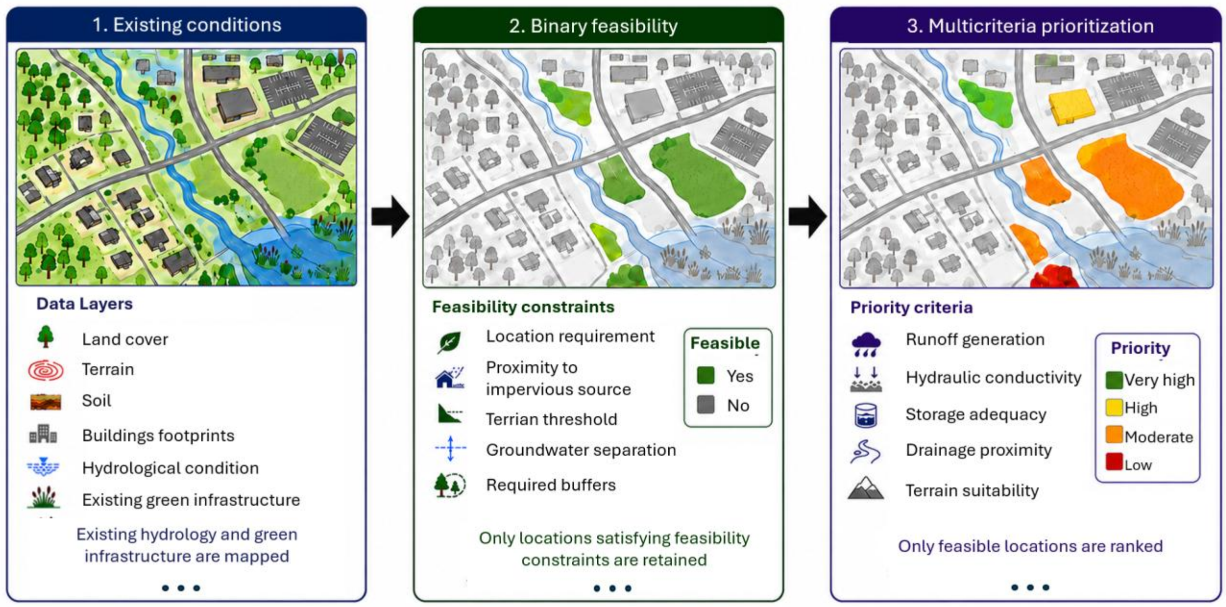

Figure 1. Conceptual structure of the PLACE framework. Existing hydrological and green infrastructure conditions are first mapped, BMP-specific feasibility constraints are then applied to delineate the feasible domain, and only feasible locations are subsequently ranked using priority criteria.

The objective of this study is to develop and demonstrate PLACE as a geospatial framework for BMP feasibility screening and prioritization. Specifically, the framework aims to (i) characterize existing hydrological conditions and GI features using high-resolution geospatial data; (ii) identify BMP-specific feasible areas using binary feasibility constraints; (iii) prioritize locations within each feasible domain using criteria representing runoff generation, hydraulic conductivity, storage, drainage, and terrain conditions; and (iv) evaluate the stability of the resulting priorities under changes in criterion weights. PLACE is intended as a screening-level decision-support framework for comparing BMP implementation opportunities and identifying candidate locations for more detailed hydrological, hydraulic, and site-specific evaluation. Its outputs represent relative runoff-retention and infiltration-support potential rather than confirmation of engineering feasibility or predictions of validated BMP performance. In Section 2, we define the decision architecture of PLACE by characterizing existing condition, binary feasibility screening and multicriteria prioritization that operationalize its three decision steps.

2. Methods

2.1. Overview

The Prioritization of Locations After Constraint Evaluation (PLACE) framework separates BMP feasibility screening from subsequent spatial prioritization. As illustrated in Figure 2, PLACE consists of three sequential steps. The first step, which is characterization of existing hydrological and green infrastructure (GI) conditions, generates the spatial information required for the two decision stages. The second step, which is binary feasibility screening, identifies the feasible domain for each BMP by retaining only locations that satisfy all applicable feasibility constraints. For any BMP b at location x, Eq. 1 defines the binary feasibility function as

$$F_b(x) = \begin{cases} 1, & if \prod_{k=1}^{K_b} C_{b,k}(x) = 1 \\ 0, & Otherwise \end{cases} \quad \text{for } b \in \{1, \dots, B\} \qquad (1)$$

where $C_{b,k}(x) \in \{0,1\}$ represents feasibility constraint k, with 1 indicating that the constraint is satisfied and 0 otherwise; $K_b$is the number of constraints applied to BMP candidate b; B is the total number of BMPs. The third step, which is multicriteria prioritization, ranks locations only within that feasible domain according to weighted priority criteria. As presented in Eq. 2, the priority is evaluated only within the feasible domain

$$P_b(x) = \begin{cases} \sum_{j=1}^{J_b} w_{b,j} S_{b,j}(x), & F_b(x) = 1 \\ NA, & F_b(x) = 0 \end{cases} \qquad (2)$$

where $P_b(x)$ is the priority index at location x under the binary feasibility locations defined by Eq. 1, J_b is number of active criteria for BMP b, $S_{b,j}(x)$ is the standardized score for criterion j, and $w_{b,j}$ is the normalized criterion weights, which sum to one. Defining priority as undefined (NA) outside the feasible domain excludes locations that fail a prerequisite feasibility constraint from prioritization rather than assigning them a low-priority score. All preprocessing, analysis and postprocessing are conducted in Python and the data and codes are available (Gebremedhin, 2026).

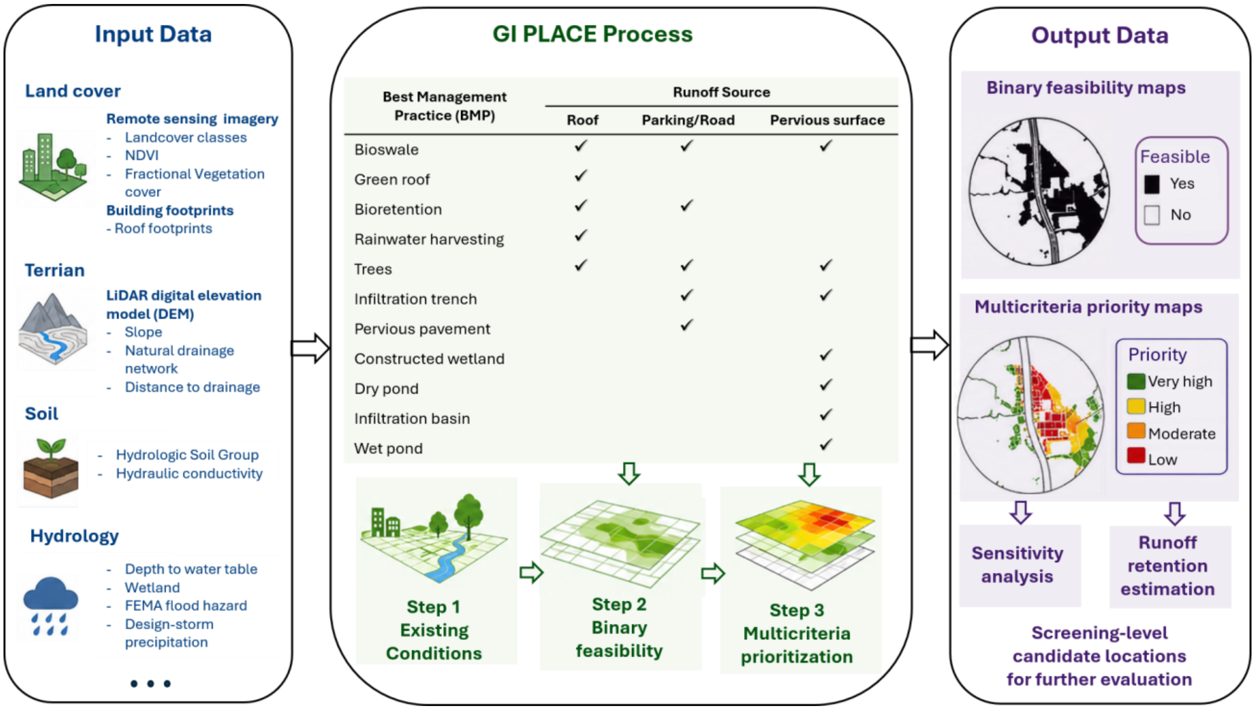

Figure 2. Method overview of the geospatial multicriteria for prioritization of GI locations after feasibility constraint evaluation (PLACE) showing input data sources and data products, green infrastructure (GI) PLACE process given three runoff sources and 11 BMPs and the output data.

2.2. Study area

The PLACE framework was demonstrated in Bonita Bay in Southwest Florida, which covers about 7.30 km^2 (Figure 3). Elevation ranges from about −0.3 to 8.0 m above mean sea level based on the 0.5 m LiDAR Digital Elevation Model, reflecting the low-relief coastal setting and limited natural drainage gradients. The area contains an interconnected stormwater management system of lakes, ponds, wetlands, pipes, inlets, weirs, and control structures (Tsegaye et al., 2024). Stormwater response is influenced by low topographic gradients, high antecedent water levels, tidal boundary effects, and shallow hydrogeologic conditions, while intense wet season rainfall and hurricanes can produce substantial flooding (Wang et al., 2024). Bonita Bay was selected to demonstrate the PLACE framework because its developed land cover, multiple runoff source types, low-relief terrain, shallow groundwater conditions, and existing water bodies create spatially variable constraints and opportunities for BMP implementation. These characteristics provide a setting for evaluating the distinction between where BMPs are feasible and how feasible locations differ in relative priority.

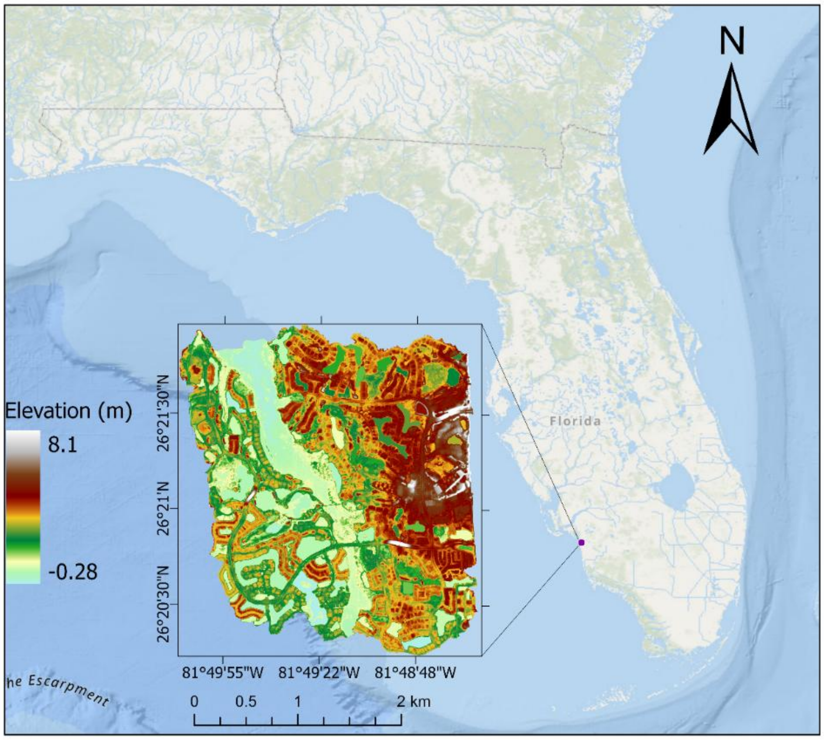

Figure 3. Location and topography of Bonita Bay case study in Southwest Florida

2.3. Data

Spatial data used to implement PLACE include land cover, terrain, soil properties, and hydrological conditions. Source datasets are processed to generate the analysis-ready geospatial products summarized in Table 1, while Table S1 identifies the function of each product and the PLACE steps in which it is used. All spatial layers are projected to a common coordinate reference system, clipped to the study area, and aligned to a common analysis grid before geospatial analysis.

Table 1. Source data and geospatial products used in the Bonita Bay demonstration of PLACE

Input data	Data source	Method	Data product
Land cover			
Multispectral aerial imagery	USDA National Agriculture Imagery Program (NAIP) (USDA FSA, 2023)	Machine-learning land-cover classification	Land cover classes
Multispectral aerial imagery	USDA National Agriculture Imagery Program (NAIP) (USDA FSA, 2023)	NDVI calculated from red and near-infrared bands	Normalized Difference Vegetation Index (NDVI)
NDVI	Derived from NAIP	Fractional vegetation cover derived as a function of NDVI	Fractional vegetation cover (FC, %)

Building footprints	Microsoft US Building Footprints (Microsoft Maps, 2018)	Projection, clipping, and spatial alignment	Building/roof footprints
Terrain			
Digital elevation model (DEM) data	2018 USGS/NRCS Lidar DEM (OCM Partners, 2026)	Terrain analysis in ArcGIS Pro	Slope
Digital elevation model (DEM) data	2018 USGS/NRCS Lidar DEM (OCM Partners, 2026)	Hydrologic terrain processing Spatial Analyst Hydrology toolset	Natural drainage network
Natural drainage network	Derived from DEM	Euclidean/proximity distance calculation	Distance to drainage
Soil properties			
Soil polygons and component attributes	USDA NRCS Web Soil Survey / SSURGO (USDA NRCS, 2024)	Spatial and tabular integration using Pandas and GeoPandas	Hydrologic soil group
Hydraulic conductivity data	U.S. Geological Survey data release (Watson & Belitz, 2024)	Projection, clipping, and alignment to analysis grid	Hydraulic conductivity
Hydrological conditions			
Depth-to-water-table data	Florida Geological Survey (FGS, 2022)	Projection, clipping, and alignment to analysis grid	Depth to water table
Wetland polygons	U.S. Fish and Wildlife Service National Wetlands Inventory (USFWS, 2026)	Projection, clipping, and spatial alignment	Wetlands
Flood-hazard polygons	FEMA Flood Hazard Services (FEMA, 2022)	Extraction of 100-year floodplain (Zone A) and spatial alignment	100-year floodplain
Design-storm precipitation grid	NOAA Atlas 14 Precipitation Frequency Data Server (National Weather Service, 2013)	Extraction and alignment of the 2-year, 24-h precipitation grid	2-year, 24-h rainfall depth
Additional application-specific data	User/application dependent	User-defined processing	Additional PLACE data products

2.3.1. Land cover

High-resolution NAIP imagery is used to classify land cover and derive vegetation-related indicators. A machine-learning workflow is implemented in a Python notebook environment to classify features. Training data are created from polygons representing buildings and parking areas/roads (impervious surfaces), trees/forest, water, and grass. A multi-feature stack is constructed from the NAIP red, green, blue, and near-infrared spectral bands, together with derived indices and spatial features, including the Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), mean visible band brightness, local texture measures represented by the standard deviations of the red and near-infrared bands, and edge magnitude derived using Sobel filtering. These additional features are used to improve class separability, particularly among spectrally similar land cover surfaces. A supervised ExtraTrees ensemble model is trained using stratified samples of labeled pixels, with 80% used for training and 20% for validation. The classifier achieved an overall validation accuracy of approximately 99%.

Fractional vegetation cover (FC) is derived as a function of the NDVI, which is calculated from high-resolution NAIP multispectral imagery. NDVI has been widely applied as an indicator of vegetation density and horizontal vegetation coverage due to its strong linear relationship with fractional vegetation cover (Zhang et al., 2013). FC can be expressed as a function of NDVI using reference values representing fully covered vegetation and bare soil conditions (Gao et al., 2020; Gebremedhin et al., 2022). As presented in Eq. 3, the FC is computed as a function of NDVI derived from NAIP,

$$FC = \frac{NDVI - NDVI_s}{NDVI_v - NDVI_s} \tag{3}$$

where NDVI represents the NDVI value of the target pixel, $NDVI_s$ represents the minimum NDVI corresponding to bare soil conditions, and $NDVI_v$ represents the maximum NDVI corresponding to dense vegetation or full vegetation cover. The lower $NDVI_s$ and upper $NDVI_v$ for vegetation were defined as the 5th and 95th percentiles, respectively (Gao et al., 2020).

The land cover and fractional vegetation cover can be derived from existing databases such as the National Land Cover Database (NLCD). However, the coarse spatial resolution of such products can reduce the accuracy of identifying and siting GI practices at the required scale. This limitation is demonstrated when comparing FC derived from NLCD with the high-resolution FC map derived from NAIP imagery (Figure 4). The NAIP-based FC more clearly represents ground-level features and spatial variability, which are critical for locating suitable sites for green infrastructure best management practices (BMPs). Features with near-zero fractional vegetation cover, such as roads, buildings, and water surfaces, are distinctly identified in the NAIP-derived FC. Similarly, vegetated cover types are mapped with sharper boundaries and improved detail, allowing for more reliable evaluation of site conditions and hydrological suitability. In contrast, these fine-scale features are generalized in the NLCD-derived fractional vegetation cover due to its coarser resolution.

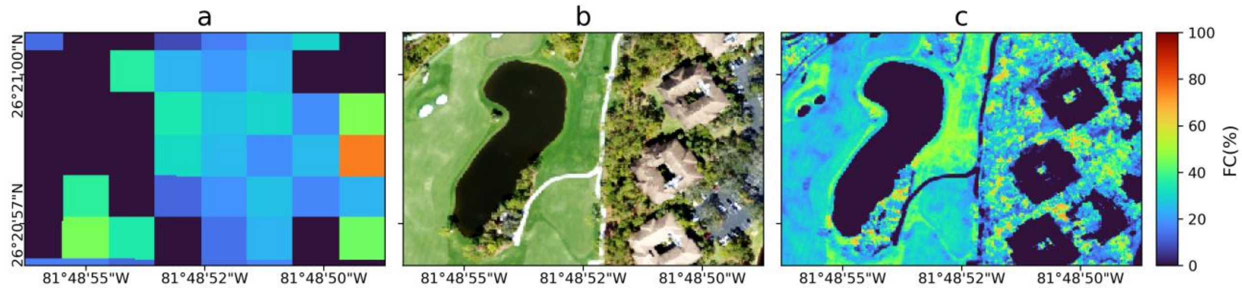

Figure 4. (a) coarse fractional vegetation cover derived from NLCD source database, (b) NAIP imagery, and (c) high-resolution fractional vegetation cover map produced from NAIP imagery with 1 m resolution.

2.3.2. Terrain

Terrain and drainage products are derived from 2018 USGS/NRCS Southwest Florida LiDAR DEM sourced through NOAA Digital Coast as shown in Table 1. The DEM is used to generate slope and natural drainage pathways. Slope is retained as an analysis layer because it contributes to both BMP-specific feasibility screening and multicriteria prioritization. The natural drainage network is derived from the LiDAR DEM using Spatial Analyst Hydrology toolset in ArcGIS Pro. The resulting drainage network is used to characterize existing drainage pathways and to derive the distance to drainage layer used in subsequent prioritization.

2.3.3. Soil

Both runoff generation and infiltration potential within the PLACE framework are derived from soil characteristics. Hydrologic Soil Groups (HSGs) are generated by linking spatial soil polygons with the corresponding SSURGO component attributes using Pandas and GeoPandas. HSG provides the soil classification used for existing condition characterization and, together with land cover, for assigning Curve Numbers in the event-based runoff calculation. Saturated hydraulic conductivity is obtained from a U.S. Geological Survey data release and used as the spatial representation of soil infiltration potential in the multicriteria prioritization.

2.3.4. Hydrology

Hydrological datasets include groundwater depth, wetlands, floodplain extent, and design storm rainfall. Data sources for depth to the water table within the Surficial Aquifer System, wetland boundaries, and the 100-year floodplain (Zone A) and rainfall are shown in Table 1. Together with the open water layer derived from NAIP, these products characterize groundwater and hydrologically sensitive conditions used in the existing condition and feasibility steps. A spatially distributed 2-year, 24-h rainfall depth is used as the design storm input for event-based runoff estimation. The rainfall product is retained as a gridded input so that spatial variability in rainfall can be propagated into runoff calculations, although rainfall variability across the Bonita Bay study area was relatively small.

2.4. PLACE process

The geospatial data products summarized in Table 1 are used as inputs to the Python-based implementation (Gebremedhin, 2026) in three sequential PLACE steps of existing condition characterization, binary feasibility screening, and multicriteria prioritization as described in this section. All raster-based analyses are performed cell-by-cell to ensure consistency among geospatial layers.

2.4.1. Existing condition characterization

This step characterizes existing hydrological and GI conditions relevant to BMP placement and subsequent prioritization. The analysis delineates riparian areas, tree/forest cover, areas designated as groundwater-recharge zones, natural drainage pathways, steep slopes, and pervious areas (Tsegaye et al., 2019). Riparian areas are generated by combining open-water areas, wetlands, and the FEMA 100-year floodplain of Zone A (buffered 30.48 m). Tree/forest areas are identified from the fractional vegetation cover raster using a threshold greater than 50%. Areas designated as groundwater-recharge zones are defined as locations corresponding to HSG A and B. Natural drainage pathways derived from the LiDAR DEM are buffered by 30.48 m, while areas with slopes greater than 25% are similarly buffered by 30.48 m. Pervious areas are delineated from non-impervious land cover after excluding riparian areas, tree/forest cover, designated recharge zones, natural drainage pathways, and steep-slope areas. These derived layers are retained as separate spatial products, and their purpose is to characterize existing conditions and establish the spatial information required by the subsequent feasibility and prioritization steps.

2.4.2. Binary feasibility screening

This step identifies a separate feasible domain for each BMP by applying BMP-specific feasibility constraints. BMPs Table 2 in are evaluated for runoff originating from roofs, parking/roads, and pervious areas. The feasibility rules presented in Table 2 are adopted from Tsegaye et al. (2019) and extended by adding the groundwater depth constraint (EPA, 2025). These rules define the prerequisite conditions for implementation, including site location, proximity to the relevant runoff source, environmental setback requirements, land slope and groundwater separation. Proximity to impervious runoff sources is quantified using the WhiteboxTools geospatial-analysis library (Lindsay, 2016). The resulting distance rasters are evaluated together with land cover, slope, groundwater-depth, and other applicable constraints using raster-based conditional operations. The feasibility screening also specifies a minimum 0.3 m separation from riparian areas for BMPs on pervious site except trees (Tsegaye et al., 2019). The output from this step is a BMP-specific binary feasibility map that is used as the spatial mask for Step 3. Users can add other feasibility rules as needed.

Table 2. BMP-specific feasibility constraints used in the PLACE for the green infrastructure best management practices

Runoff source	Green infrastructure BMP	Location requirement	Proximity to impervious source	Slope	Groundwater depth
Roof	Bioswale	Pervious area adjacent to proposed or existing building	> 3.04 m and < 15.24 m	< 5%	≥ 0.61 m
	Green roof	Proposed or existing building (at source)	At source	n/a	n/a
	Bioretention (Rain Garden)	Pervious area adjacent to proposed or existing building	> 3.04 m and < 15.24 m	< 5%	≥ 0.61 m
	Rainwater harvesting	Proposed or existing building (at source)	At source	n/a	n/a
	Trees	Pervious area adjacent to proposed or existing building	> 3.04 m and < 15.24 m	All slopes	n/a
Parking/road	Infiltration trench	Pervious area adjacent to proposed or existing parking lot/road	> 0.30 m and < 15.24 m	< 5%	≥1.22 m
	Bioswale	Pervious area adjacent to proposed or existing parking lot/road	> 0.30 m and < 15.24 m	< 5%	≥ 0.61 m
	Pervious pavement	Proposed or existing parking lot/road (at source)	At source	n/a	n/a
	Bioretention (Rain Garden)	Pervious area adjacent to proposed or existing parking lot/road	> 0.30 m and < 15.24 m	< 5%	≥ 0.61 m
	Trees	Pervious area adjacent to proposed or existing parking lot/road	< 15.24 m	< 5%	n/a
Pervious surfaces on site	Bioswale	Pervious area on site	> 15.24 m	< 5%	≥ 0.61 m
	Constructed wetland	Pervious area on site	> 15.24 m	< 15%	≥ 1.22 m
	Dry pond	Pervious area on site	> 15.24 m	< 15%	>1.22 m
	Infiltration basin	Pervious area on site	> 15.24 m	< 5%	≥ 1.22 m
	Trees	Pervious area on site	> 15.24 m	< 5%	n/a
	Wet pond	Pervious area on site	> 15.24 m	< 15%	>1.22 m

2.4.3. Multicriteria prioritization

This step rank locations within each BMP-specific feasible domain according to priority criteria shown in Table 3. The criteria are selected to represent runoff management demand, soil infiltration conditions, storage relative to runoff, separation from natural drainage pathways, and terrain conditions. Nonapplicable criteria are excluded, and the weights of the remaining criteria are renormalized before calculation of the priority index. For example, saturated hydraulic conductivity is not applied to green roofs. These criteria are selected for this case study demonstration, while users may modify the BMP-specific criteria or add others according to local conditions and planning objectives.

Spatially distributed runoff depth is estimated using the USDA Curve Number (CN) method (USDA, 1986). Land cover is separated into impervious and pervious surfaces, with pervious areas further differentiated by HSG. Curve Numbers are assigned according to land cover and soil group classifications, and a spatially distributed 2-year, 24-h rainfall event. The CN calculation is used to represent the spatial distribution of event-based runoff generation rather than to simulate routing of runoff from individual source areas to BMP locations. The runoff depth is calculated using Eq. 4

$$Q_i = \begin{cases} \dfrac{\left(P - I_{a,i}\right)^2}{P - I_{a,i} + S_i}, & \text{if } P > I_{a,i} \\ 0, & \text{if } P \leq I_{a,i} \end{cases} \tag{4}$$

where Q_i is runoff depth at cell i, P is rainfall depth in inches, $I_{a,i} = 0.2S_i$ is the initial abstraction, and S_i is the potential maximum retention after runoff begins, presented in Eq. 5.

$$S_i = \frac{1000}{\mathrm{CN}_i} - 10 \tag{5}$$

Runoff generation potential was represented by the CN-based runoff-depth raster, consistent with the use of spatial runoff indicators in GIS-based multicriteria analysis (Al-Ghobari & Dewidar, 2021). The runoff-depth values were standardized to a 0–1 scale using min–max normalization prior to multicriteria aggregation (Krise et al., 2026). This transformation expresses the relative spatial variation in normalized runoff generation potential and makes the runoff criterion comparable with other criteria measured in different units. Because rainfall is nearly uniform across the case study, the normalized variation primarily reflects differences in land cover and hydrologic soil conditions represented by the Curve Number. Storage adequacy can be defined as the ratio of captured runoff to generated runoff volume, following runoff retention and volume capture metrics for stormwater BMP assessment (Clar et al., 2004). This evaluates the capacity of a feasible BMP. At cells where multiple BMP feasibility domains overlap, storage capacity is represented by the maximum capacity of a single feasible BMP rather than the sum of their capacities. Saturated hydraulic conductivity is an indicator of soil water-transmission capacity, which can be classified using thresholds from USDA

NRCS HSG criteria (USDA NRCS, 2025). Highly permeable pervious areas are mapped separately as potential groundwater recharge zones and excluded from the residual pervious layer during existing condition characterization. However, this characterization does not necessarily make these areas unavailable for BMP siting because they may be included as candidate locations where sufficient area is available and applicable feasible requirements are satisfied. Distance to natural drainage is represented by the distance from each cell to the nearest natural drainage pathway, consistent with GIS-based stormwater BMP siting approaches that use distance from streams as a spatial criterion (Saadat Foomani & Malekmohammadi, 2020; Yimer et al., 2024).

Table 3. Priority criterion classification used in the multicriteria prioritization within feasible BMP sites.

Criterion	High criterion suitability	Moderate criterion suitability	Low criterion suitability
Runoff generation potential (%)	>75	50–75	<50
Saturated hydraulic conductivity (inch/hr)	>5.67	1.42-5.67	<1.42
Storage adequacy (%)	>75	50-75	<50
Distance to natural drainage (m)	>100	75-100	30-75
Slope suitability (%)	<2	2–5	>5

The criterion layers are transformed into a common continuous scale before weighting so that variables with different units can be combined within the priority index. The resulting standardized criterion layers are evaluated only within the corresponding BMP-specific feasible domain. Criterion weights are derived using the Analytic Hierarchy Process (AHP) and Saaty's pairwise-comparison scale (Saaty, 2008). Pairwise comparison values express the relative importance of the criteria. As shown in Table 4, runoff generation is judged equally important as saturated hydraulic conductivity (1), slightly more important than storage adequacy (2), moderately more important than distance to natural drainage (3), and strongly more important than slope suitability (5). The principal eigenvector of the pairwise comparison matrix is then calculated to obtain normalized weights, and logical consistency is evaluated using the consistency ratio (CR), with CR < 0.1 considered acceptable (Saaty, 1987). The resulting matrix had a CR of 0.006, indicating internally consistent pairwise judgments. Within PLACE, additional criteria may be incorporated for other planning objectives, in which case the pairwise comparisons, normalized weights, and consistency ratio would be recalculated accordingly. A consistent AHP weighting structure is applied across BMPs to maintain a common decision framework for their respective feasible domains. For each raster cell within the BMP-specific feasible domain, a composite priority index is calculated using the weighted linear combination in Eq. 2.

Table 4. Pairwise comparison matrix and normalized criterion weights derived using the Analytic Hierarchy Process for PLACE multicriteria prioritization.

Criterion	Runoff generation potential	Saturated hydraulic conductivity	Storage adequacy	Distance to natural drainage	Slope suitability	Normalized weight
Runoff generation potential	1	1	2	3	5	0.33
Saturated hydraulic conductivity	1	1	2	3	4	0.32
Storage adequacy	1/2	1/2	1	2	3	0.18
Distance to natural drainage	1/3	1/3	1/2	1	2	0.11
Slope suitability	1/5	1/4	1/3	1/2	1	0.06

For priority classification, the index values are classified separately for each BMP using the 25th and 75th percentile thresholds. Values above 75th percentile are assigned to the primary priority class, values between 25th percentile and 75th percentile to the secondary priority class, and values below 25th percentile to the tertiary priority class. Quantile classification is used because the objective is to rank locations and no empirically validated index thresholds are available to define absolute BMP performance. In the PLACE framework, users may adjust criterion weights and priority classification to reflect planning objectives, implementation capacity, or the desired degree of spatial selectivity.

2.5. Diagnostics and sensitivity analysis

Pearson correlation and principal component analysis (PCA) are used to evaluate relationships and potential redundancy among the criteria included in the multicriteria prioritization. Pearson correlation coefficients are first calculated for each BMP using criterion values within its BMP-specific feasible domain to evaluate pairwise associations. PCA is subsequently applied to the standardized continuous criterion values using the PCA function in the scikit-learn library. Standardization of priority criteria (Table 3) is performed prior to PCA to minimize the influence of differences in criterion scales. The percentage of variance explained by each principal component is used to evaluate the dimensional structure of the criterion set and the extent to which the criteria represent overlapping or complementary spatial information (Jolliffe & Cadima, 2016).

A sensitivity analysis is conducted to evaluate the stability of the multicriteria prioritization under uncertainty in the AHP criterion weights. Each criterion weight as shown in Table 4 is independently perturbed by ±10% from its baseline value, while the remaining weights are proportionally rescaled so that the normalized weights continue to sum to one. For each perturbation scenario, the priority index is recalculated. Sensitivity is evaluated using both changes in priority class area and spatial agreement with the baseline classification. For each BMP and perturbed criterion, the area assigned to each priority class is compared with the corresponding baseline area, and a spatial overlap ratio is calculated as the proportion of baseline class cells that remain in the same class after perturbation. The overlap ratio is expressed as defined in Eq. 6.

$$O_c = \frac{|B_c \cap P_c|}{|B_c|} \tag{6}$$

where B_c is the set of cells assigned to class c under the baseline weights and P_c is the corresponding set under the perturbed weights. Values approaching 1 indicate high spatial stability, whereas lower values indicate greater sensitivity of the priority pattern to the perturbed criterion.

2.6. Screening-level runoff retention estimation

In addition to the spatial prioritization, which in general represents relative runoff retention and infiltration support potential of each BMP, screening-level runoff retention potential is estimated for each runoff source. Runoff volume is calculated by multiplying the CN-based runoff depth by cell area and aggregating the resulting volumes for each runoff source category. BMP storage capacity is estimated independently by multiplying the assigned effective storage depth for each BMP by the area of each feasible cell and summing across its complete feasible domain. Effective storage depths were adopted from Tsegaye et al. (2019). For each BMP, potential runoff retention is calculated as the lesser of total candidate storage capacity and runoff generated by the corresponding source category. Each BMP is evaluated independently, assuming implementation across 100% of its mapped feasible domain. Because feasible domains may overlap and multiple BMPs cannot necessarily occupy the same location, the resulting values represent independent maximum screening scenarios rather than additive BMP siting.

3. Results

3.1. Existing-condition characterization

The land cover and terrain characterization showed a predominantly vegetated, low-relief landscape (Figure 5). The slope pattern indicates predominantly low-gradient terrain, with most areas below 5% and only localized steeper gradients near drainage features, which are captured by the high-resolution lidar DEM (Figure 5a). Trees/forest represented approximately 45% of the study area and constituted the dominant land cover class as illustrated in Figure 5b. Impervious surfaces accounted for approximately 35% that include buildings, roads, parking areas, and other developed impervious surfaces. Grass and open space represented approximately 11%, while open water bodies accounted for about 9% as shown in Figure 5b. The distributions of NDVI (Figure 5c) and fractional vegetation cover (Figure 5d) are consistent with the classified land cover. Moderate to high NDVI values occurred primarily in forested and grass areas, whereas low values are associated with impervious surfaces and water. Fractional vegetation cover further resolved concentrated vegetation within residential areas and along green corridors.

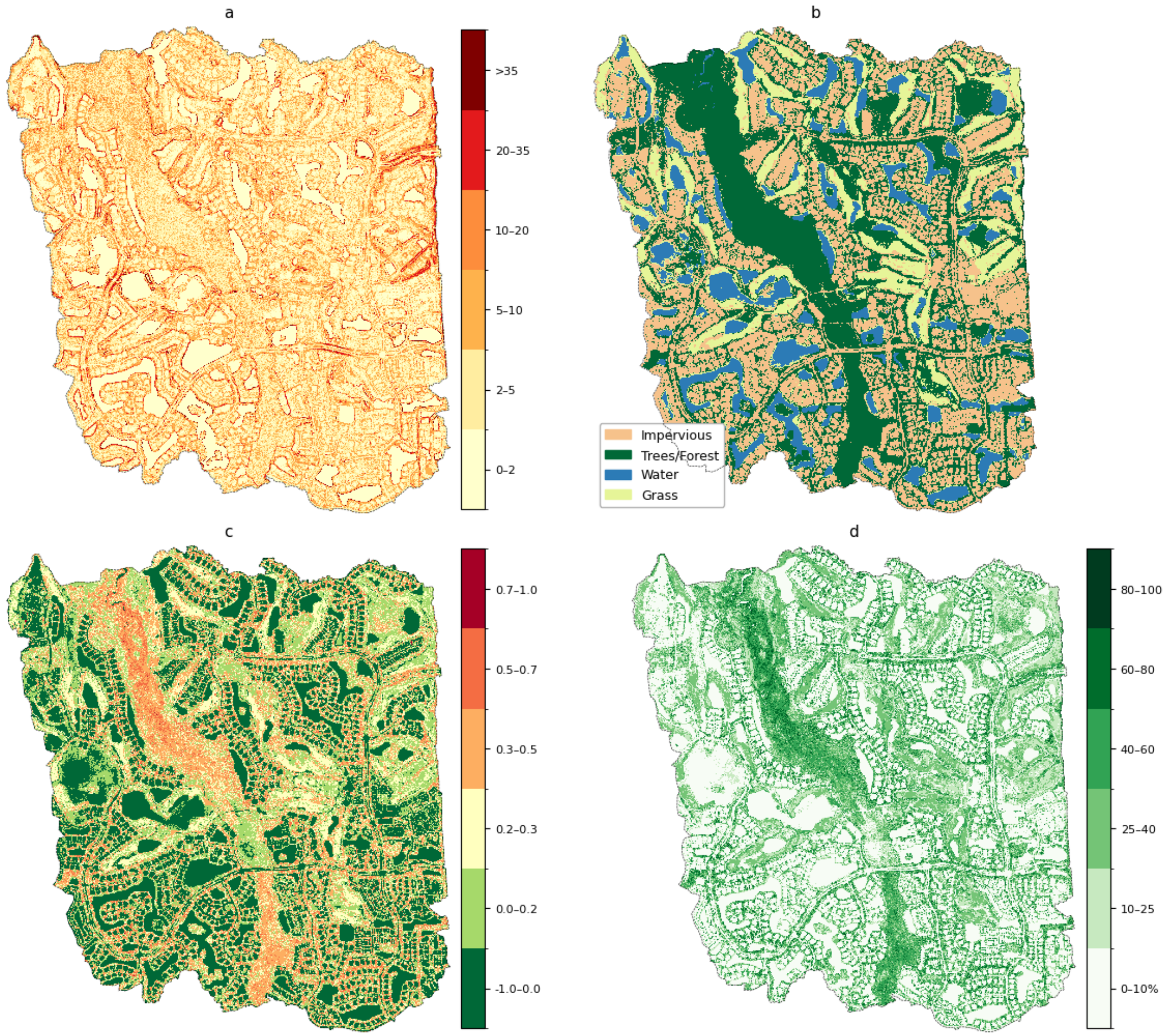

Figure 5. Topographic, land cover and biophysical characteristics of Bonita Bay: (a) slope (%), (b) land cover classification, (c) Normalized Difference Vegetation Index (NDVI), and (d) fractional vegetation cover (%).

The derived hydrological and GI layers demonstrate the spatial overlap between developed land, existing green infrastructure, and hydrologically sensitive areas as shown in Figure 6. Approximately 65% of the study area is non-impervious. Riparian areas, defined from wetlands, open water, the 100-year floodplain, and the associated buffer, cover about 30% of the study area. Buffered natural drainage pathways accounted for about 15% and formed a connected network through residential and open space areas. Areas designated as groundwater recharge zones based on HSG A and B represent about 18% of the case study area. Steep slope areas represent less than 2% of the study area, reflecting low-relief terrain. About 20% of the study area consisted of pervious land that did not overlap with riparian areas, tree/forest cover, potential groundwater recharge zones, buffered natural drainage pathways, or steep slopes. This residual pervious land is retained as the candidate pervious domain for subsequent binary feasibility screening of each BMP (Step 2).

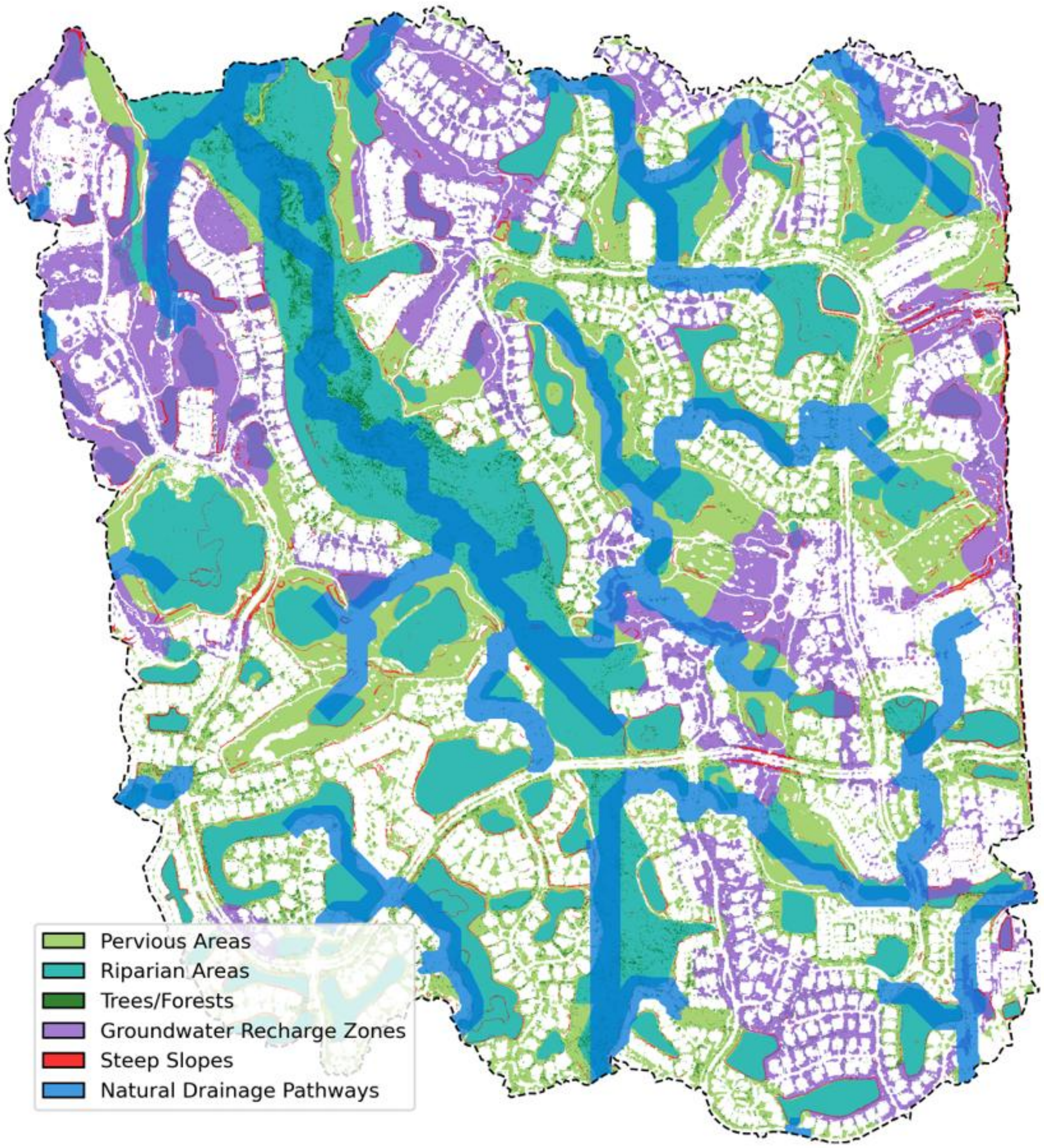

Figure 6. Spatial distribution of existing hydrological and green infrastructure features, including residual pervious areas, riparian areas, tree/forest cover, areas designated as groundwater-recharge zones, steep slopes, and natural drainage pathways. The layers are displayed separately because their spatial domains may overlap.

3.2. Binary feasibility screening

Binary feasibility screening produced a distinct domain for each of the BMPs. Figure 7 shows results for green-roof BMP for roof runoff source, pervious pavement and bioswale BMPs for parking/road runoff source, and trees BMP for pervious-surface runoff source. The binary feasibility maps for the remaining 12 BMPs are provided in Supplementary material Figure S1. The extent and spatial configuration of the feasible domains vary according to the constraints in Table 2. Feasible locations are generally concentrated near developed runoff source areas, whereas feasible areas for BMPs within pervious surfaces are more spatially fragmented. Across the three runoff sources of roof, parking/road, and pervious surfaces, the resulting feasible domains reflect the interaction between runoff source geometry and BMP-specific constraints (Figure 7 and Figure S1). Feasible-domain extent and spatial continuity varied systematically with BMP siting, with at-source practices generally covering broader domains than practices dependent on adjacent or residual pervious area.

Green roof feasibility is confined to building footprints distributed throughout the developed areas (Figure 7a) because it is at-source practice whose feasibility determined primarily by the presence of building footprints and not constrained by local groundwater depth or slope. Similarly, pervious pavement show an extensive and spatially connected feasible domain along the parking/road network (Figure 7b). In contrast, BMPs requiring adjacent pervious area produce narrower and less continuous domains. For example, bioswales in parking/road occur primarily as patches along impervious area margins after the applicable constraints are applied (Figure 7c). Pervious-surface BMPs are the most spatially restricted, with feasible tree locations occurring as sparse and discontinuous patches (Figure 7d) because only a limited residual pervious area remained after existing hydrological and GI features and BMP-specific constraints are considered. Larger storage-oriented practices, including constructed wetlands, infiltration basins, dry ponds, and wet ponds, were even more limited in spatial extent (Figure S1). The results show that feasible-domain extent and connectivity are controlled not only by runoff-source geometry but also by whether a BMP could be sited at source or required suitable adjacent or residual pervious area.

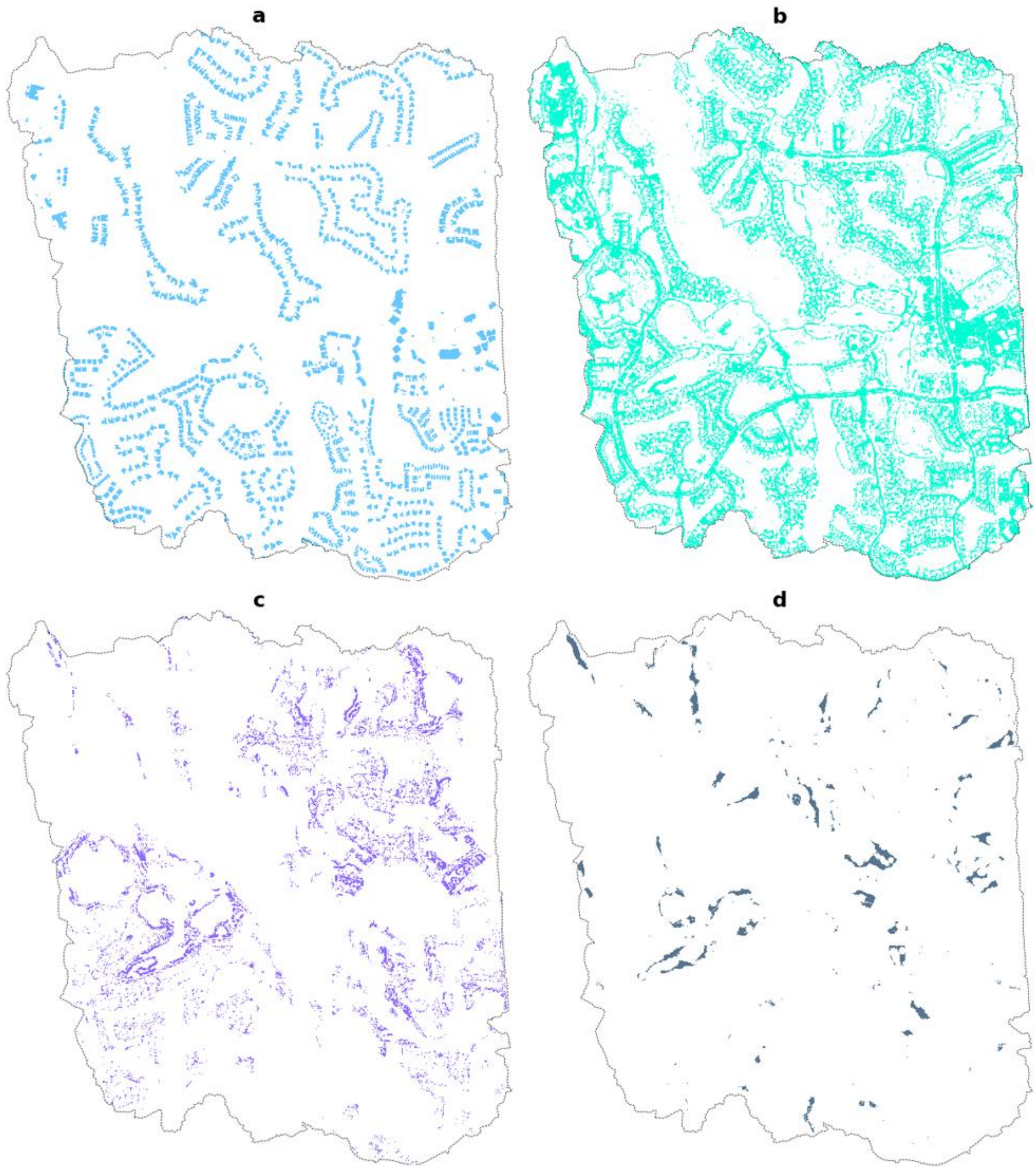

Figure 7. BMP-specific feasible domains identified through binary feasibility screening using the feasibility constraints in Table 2: (a) green roof for roof runoff sources, (b) pervious pavement for parking/road runoff sources, (c) bioswale for parking/road runoff sources, and (d) trees for pervious surfaces runoff sources.

3.3. Multicriteria prioritization

Multicriteria prioritization differentiated spatial priority within the BMP-specific feasible domains identified by the preceding feasibility screening. The resulting priority index is classified into primary, secondary, and tertiary zones to represent relative ranking within each feasible domain. Representative priority maps corresponding to the BMPs shown in Figure 7 are presented in Figure 8, while the remaining BMP priority maps are provided in Supplementary Figure S2.

The spatial configuration of priority varies considerably among BMPs. Green roofs show spatial differentiation among building footprints, with higher-priority locations occurring on selected portions of the otherwise broadly distributed feasible roof domain (Figure 8a). Pervious pavement similarly retains a broad and spatially connected feasible domain, but priority varies throughout the parking/road domain, producing distinct higher- and lower-priority segments within the same siting surface (Figure 8b). In contrast, bioswales for parking/road exhibit more fragmented priority patterns along impervious-source margins (Figure 8c), while pervious surface trees showed localized priority patches within an already discontinuous feasible domain (Figure 8d). Other pervious-surface BMPs exhibit similarly localized patterns because their feasible domains are comparatively restricted (Figure S2). The prioritization results generally show that broad feasible domains can contain large internal spatial variation, whereas more constrained BMPs tend to concentrate priority within smaller, more fragmented locations.

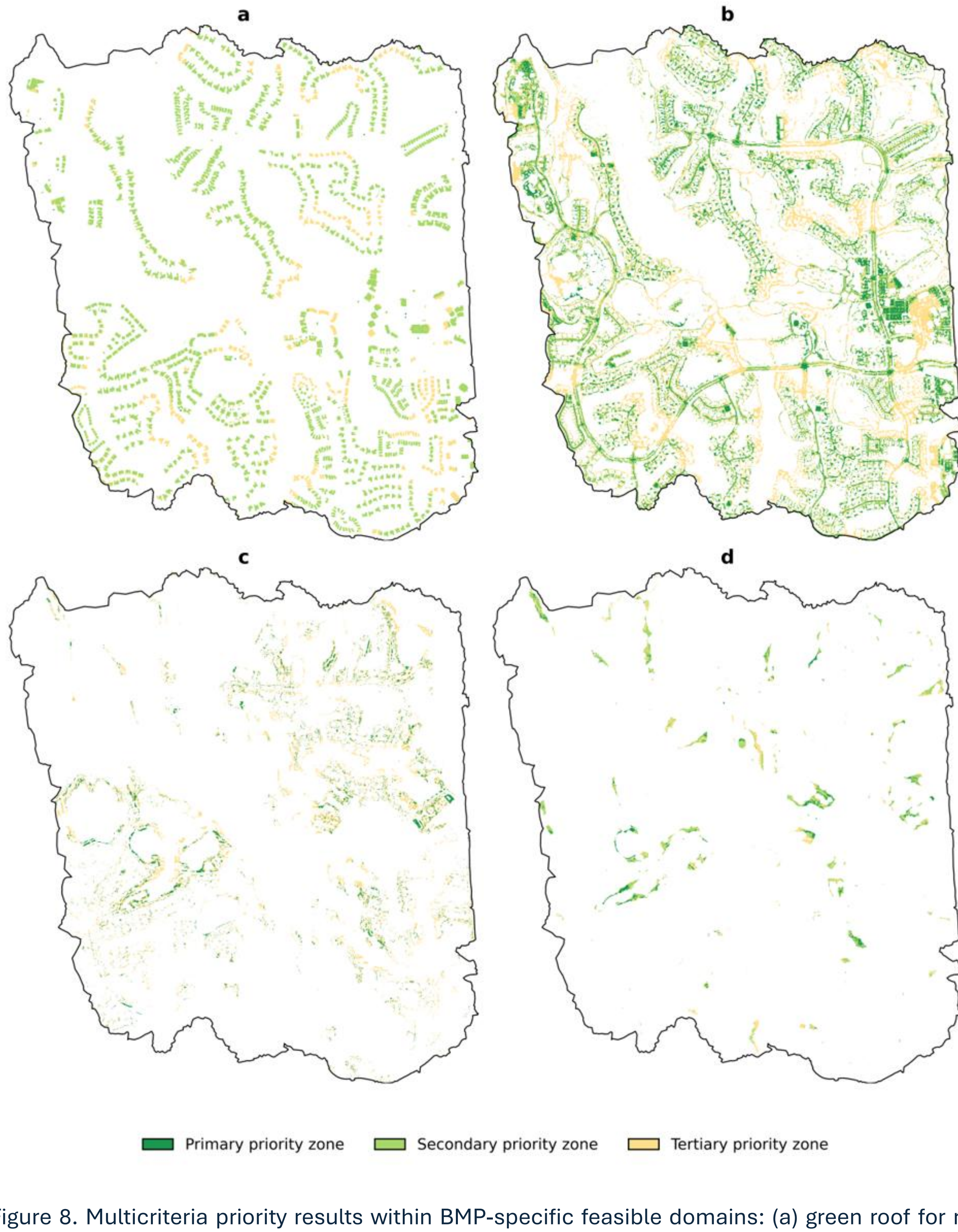

Figure 8. Multicriteria priority results within BMP-specific feasible domains: (a) green roof for roof runoff sources, (b) pervious pavement for parking/road runoff sources, (c) bioswale for parking/road runoff sources, and (d) trees for pervious surfaces runoff sources.

3.4. Diagnostics and sensitivity analysis

The diagnostic analyses show that the priority criteria contribute distinct spatial information within each BMP feasible domain. Pearson correlations among the priority criteria are generally weak, indicating limited pairwise redundancy within the BMP-specific feasible domains (Figure S3). The highest BMP-specific negative correlation is shown for the bioswale for parking/road runoff source, particularly between runoff generation potential and storage adequacy. PCA shows a similar pattern, with the first principal component explaining 25.9% of the variance and the first four components together explaining 85.5% (Figure S4). These results indicate that the priority criterion set is not dominated by a single spatial pattern.

The sensitivity analysis shows generally limited changes in priority patterns under the ±10% weight perturbations (Figure 9). Most BMPs exhibited small changes in priority class and high spatial overlap with the baseline classification. The largest responses occur for parking/road BMPs, particularly infiltration trenches, which show greater class-area changes and lower overlap when criterion weights are perturbed. Parking/road trees are also comparatively sensitive to storage adequacy and distance to natural drainage, whereas parking/road bioswales and bioretention/rain gardens show smaller responses. Roof and pervious surface BMPs are less sensitive under the evaluated perturbations. Across criteria, distance to natural drainage and storage adequacy produce the clearest changes for several BMPs, while saturated hydraulic conductivity produces the smallest responses. The priority patterns remained generally stable under the range of weight perturbations.

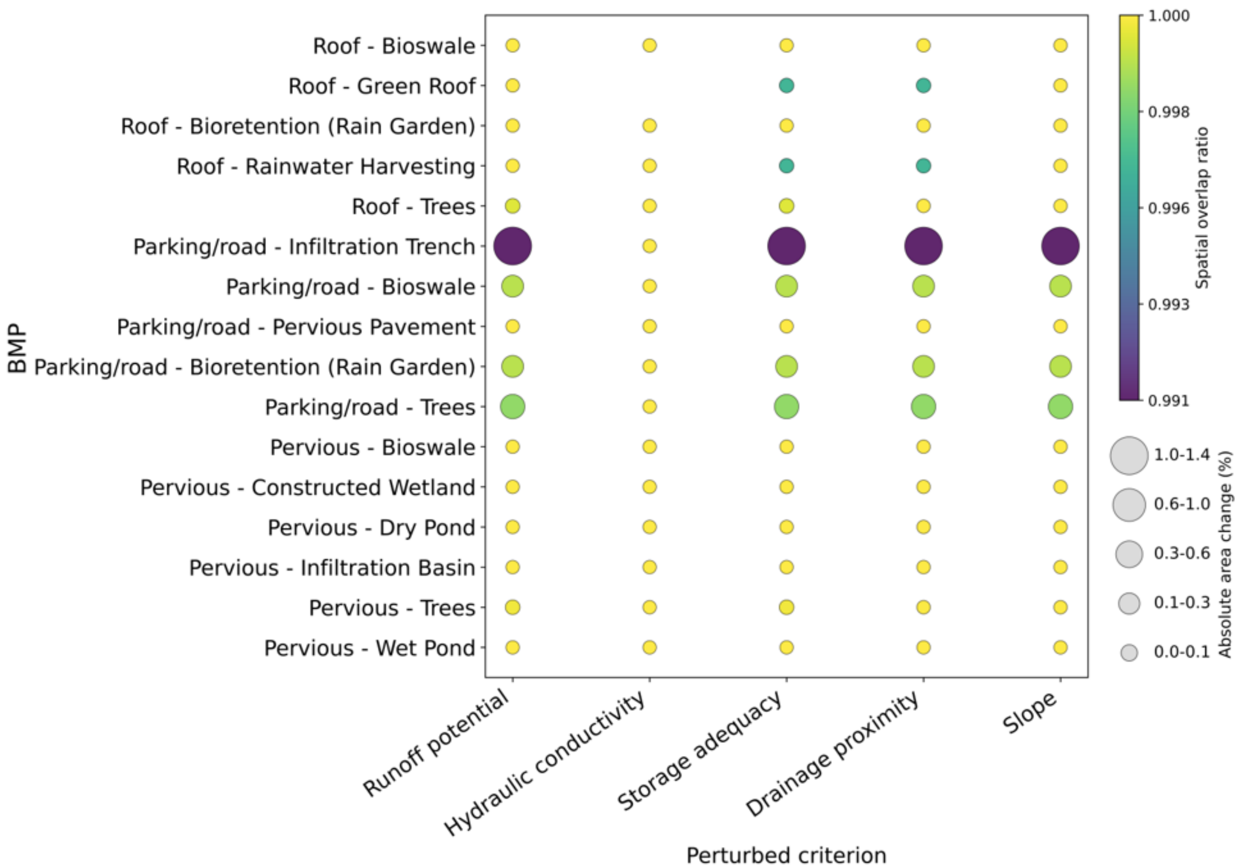

Figure 9. Sensitivity of BMP priority classes to ±10% one-at-a-time criterion weight perturbations. Bubble size represents the absolute change in priority class area, and color represents the spatial overlap ratio between the corresponding baseline and perturbed priority classes.

3.5. Screening-level runoff retention

Screening-level storage-based runoff-retention potential differed among runoff sources and BMP sitings (Table 5). Parking/road BMPs show the greatest retention potential, with pervious pavement reaching approximately 98% of parking/road source runoff under the assumed maximum implementation scenario. Roof-based BMPs show an intermediate pattern, with at-source practices such as green roofs and rainwater harvesting providing greater storage-based retention potential than practices dependent on adjacent pervious land. In contrast, pervious-surface BMPs generally show lower retention potential, with trees and bioswales exceeding constructed wetlands, ponds, and infiltration basins because the latter are restricted to much smaller feasible domains. The estimated retention potential reflects the combined influence of source runoff, mapped feasible area, and assumed BMP storage depth. Accordingly, this should be interpreted as a screening-level storage scenario rather than simulated hydraulic connectivity or event-scale BMP performance.

Table 5. Maximum storage based runoff retention potential of each BMP assuming siting across 100% of each mapped feasible domain

Runoff source and BMPs	Runoff source (m^3)	BMP area (m^2)	Estimated runoff retention (m^3)	Runoff retention relative to runoff source (%)
Roof runoff source	89571			
Bioswale		97832	7394	8.3
Green roof		791826	51106	57.1
Bioretention (rain garden)		97832	7394	8.3
Rainwater harvesting		791826	51106	57.1
Trees		306786	14009	15.7
Parking/road runoff source	216899			
Infiltration trench		43569	3263	1.5
Bioswale		377901	28209	13.0
Pervious pavement		1937397	212617	98.0
Bioretention (rain garden)		377901	28209	13.0
Trees		508978	33313	15.4
Pervious runoff source	83175			
Bioswale		82733	6159	7.4
Constructed wetland		4939	372	0.4
Dry pond		4939	372	0.4
Infiltration basin		3788	285	0.4
Trees		142009	7706	9.3
Wet pond		4939	372	0.4

4. Discussion

4.1. Site-specific BMP siting

The comparison among BMPs in Bonita Bay shows that siting opportunities are controlled by how each BMP relates to its runoff source and the amount and condition of land available for placement. First, at-source BMPs are mainly controlled by the extent of roof or paved surfaces. Practices such as pervious pavement, green roofs, and rainwater harvesting can be sited directly on the corresponding runoff-generating surface and therefore depend primarily on the spatial extent of those surfaces. Second, BMPs requiring adjacent pervious areas have additional proximity, groundwater, and terrain constraints. These practices are limited not only by runoff-source geometry but also by the availability of nearby pervious areas that satisfy the applicable siting requirements. This observation is consistent with the role of GI in managing runoff close to where it is generated through storage, infiltration, and retention processes (Ahiablame et al., 2012; Li et al., 2019; Öztürk et al., 2024). Third, BMPs dependent on residual pervious surface are the most restricted because that land is fragmented or already associated with other hydrological functions. Although developed landscapes may contain substantial pervious area, portions of that land may overlap riparian areas, drainage pathways, existing vegetation, groundwater-related conditions, or other constraints. In other words, knowing how much pervious land exists does not indicate how much of it is available for BMP siting. This pattern is consistent with previous studies showing that GI planning depends on the spatial arrangement of candidate sites and their relationship to local physical and hydrological conditions (Baldwin et al., 2022; Martin-Mikle et al., 2015). Collectively, these results show that PLACE can inform how runoff-source geometry, land availability, and local hydrological constraints jointly shape BMP siting.

4.2. Implications and transferability

The practical implication of the framework is that it provides a structured basis for progressively refining BMP siting alternatives. First, feasible area alone is insufficient because priority varies within it. In the Bonita Bay demonstration, relatively extensive feasible domains often show spatial variation in priority, indicating that delineating feasible areas alone provides limited guidance on where subsequent evaluation should be directed. This is consistent with studies showing that GI outcomes depend strongly on spatial placement and local hydrological conditions (Adhikari et al., 2024; J. Wang et al., 2023). Second, the framework helps focus subsequent detailed evaluation while retaining lower-priority feasible alternatives. Hydrological, hydraulic, engineering, or site-specific analyses can be directed toward higher-priority locations without treating lower-priority locations as infeasible. Third, priorities can be reassessed as objectives, criteria, or weights change. Changes in hydrological criteria, costs, management objectives, or weighting structures may alter the relative ranking of feasible locations without necessarily redefining the feasible domain (Khodadad et al., 2025; Tansar et al., 2023). These characteristics allow this framework to progressively refine spatial alternatives while retaining flexibility in the decision process, which supports its transferability to other study areas.

The transferability of PLACE framework is based on retaining its decision structure while adapting its inputs and parameterization to local conditions. First, data sources and resolution can be adapted. In Bonita Bay, high-resolution aerial imagery and LiDAR were important for resolving roofs, roads, narrow pervious areas, vegetation, and drainage features, whereas different datasets or resolutions may be appropriate at other spatial scales. Second, BMPs, criterion thresholds, weights, rainfall, and hydrological representations can be adapted. These components can be modified according to the physical conditions of study area, available data, BMP requirements, and planning objectives. Third, additional decision criteria can be incorporated that may include pollutant removal, implementation cost, maintenance requirements, ecological benefits, or stakeholder preferences when relevant to the planning problem. The framework can be transferred across study areas through local adaptation of its data and decision components, although its output remains at screening-level as discussed in the following section.

4.3. Limitations

PLACE is intended for screening level, and its mapped output should be interpreted as relative planning information rather than predictions of actual BMP performance. The Curve Number method estimates event-based spatial runoff generation (USDA, 1986), but it does not simulate routing from runoff source cells to candidate BMPs, drainage system hydraulics, or time-varying interactions among runoff generation, storage, infiltration, and discharge. Therefore, the mapped relationships between runoff sources and BMP locations represent spatial screening opportunities rather than verified hydraulic connectivity. The runoff retention estimates are likewise conditional on the selected design storm, assumed BMP storage depths, mapped feasible area, and siting across the full feasible domain. Each BMP is evaluated independently even where feasible domains overlap or multiple practices could target the same runoff source. These estimates represent maximum storage-based scenarios for individual BMPs. Actual event retention would also depend on BMP geometry and design, antecedent conditions, infiltration during the event, drainage behavior, and interactions among practices.

Additional uncertainty arises from the spatial representation and multicriteria formulation. Raster-based feasibility screening does not explicitly account for parcel ownership, constructability, minimum contiguous area, or facility geometry, which is particularly important for larger practices such as wetlands, ponds, and infiltration basins. Priority rankings are also conditional on the selected criteria and AHP weights. Although criterion diagnostics showed limited redundancy and the ±10% weight perturbations produced relatively small changes for most BMPs, this analysis does not capture uncertainty associated with feasibility thresholds, criterion selection, or alternative planning objectives. In addition, the percentile-based priority classes are relative to each BMP-specific feasible domain. Hence, a primary priority location represents a higher relative rank within that BMP scenario rather than an absolute level of hydrological benefit or direct equivalence with primary priority locations for other BMPs.

Runoff retention, infiltration potential, and groundwater recharge should also be interpreted separately. Urban imperviousness can reduce natural infiltration and recharge pathways (Pasquier et al., 2022), while spatial indicators can identify locations with conditions favorable for infiltration-oriented planning (Šiljeg et al., 2023). However, runoff retention does not necessarily result in groundwater recharge. For example, green roofs can retain rainfall without transferring that water directly to the aquifer. Also, this framework does not simulate or quantify recharge. Therefore, areas identified from HSG A and B during the first step should be considered areas with soil conditions suitable for infiltration and potential recharge.

5. Conclusions

This study developed the Prioritization of Locations After Constraint Evaluation (PLACE) framework to support spatial screening of green infrastructure BMPs for stormwater management. PLACE organizes the analysis through existing-condition characterization, binary feasibility screening, and multicriteria prioritization within the resulting feasible domains. Its methodological contribution is the explicit distinction between locations that fail required siting conditions and locations that remain feasible but differ in relative priority for different BMPs. This distinction allows feasible alternatives to be retained while directing subsequent evaluation toward locations that better satisfy the selected planning criteria. Thus, PLACE provides a decision architecture for progressively refining spatial alternatives rather than collapsing feasibility and priority into a single composite suitability score.

The Bonita Bay application demonstrated how PLACE differentiates BMP siting opportunities, spatial priorities, and screening-level runoff-retention potential. First, BMP siting opportunities depended strongly on the relationship between runoff-source configuration, siting requirements, and available receiving land. At-source practices associated with roofs and parking/roads generally retained broader feasible domains, whereas BMPs requiring adjacent or residual pervious area were more restricted by fragmented land and local hydrological constraints. Second, feasible area alone did not indicate where subsequent evaluation should be concentrated because priority varied substantially within some feasible domains. Multicriteria prioritization identified spatial differences among locations that had already satisfied the required siting conditions, with priority patterns remaining generally stable under the tested ±10% perturbations of criterion weights. Third, screening-level runoff-retention potential reflected the interaction among runoff-source volume, feasible-domain extent, and assumed BMP storage. Pervious pavement provided the greatest storage-based retention potential for parking/road runoff under the assumed maximum siting scenario and green roofs and rainwater harvesting provided comparatively high runoff retention for roof runoff source, while most pervious-surface BMPs were more limited by their smaller feasible domains. Collectively, these results demonstrate that PLACE can reveal how landscape configuration and local constraints shape both BMP siting opportunities and their relative spatial priority.

The PLACE framework is an adaptable screening framework for early-stage BMP planning. As an adaptable framework, its constraints, criteria, weights, hydrological inputs, and spatial resolution can be modified to reflect local conditions, available data, and planning objectives. As a screening framework, its outputs should not be interpreted as predictions of validated BMP performance or confirmation of engineering feasibility. Rather, the resulting feasible and prioritized locations provide a focused basis for subsequent hydrological, hydraulic, groundwater, engineering, economic, ecological, or field-based evaluation. Thus, PLACE helps narrow and prioritize the locations that should be considered for more detailed BMP evaluation and design.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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AI assistance statement

Coding for data preprocessing, post-processing and plotting were performed using Python with instructed assistance from ChatGPT. AI models were also used to improve text clarity, succinctness, logical flow, and overall polish. AI contributions were verified for accuracy and relevance.

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